



User Guide

Table of Contents

1. Installation
 - Mac Installation
 - PC Installation
2. Quick Start
3. User Interface
 - General Settings
 - Processing Options
4. Exporting Files
5. Supported File Types
6. Technical Details
7. Troubleshooting
8. Support

Installation

WavFix is available as both an .app for Mac and an .exe for Windows. WavFix is not signed by Apple, so in order to run it on Mac, you may need to allow it to be opened in System Preferences.

Mac Installation:

1. Locate the downloaded .zip file and extract it.
2. Double-click the extracted WavFix application to run it.
3. If the app does not open, go to Privacy & Security settings in System Preferences, scroll down to "Security" and you should see "WavFix was blocked because it is from an unidentified developer."
4. Click "Open Anyway"
5. The application will run without you doing this again in the future.
6. WavFix includes a built-in high-quality audio conversion engine. Advanced users may optionally choose FFmpeg as an external conversion backend in Processing Options; FFmpeg is free but must be installed separately.

PC Installation:

1. Locate the downloaded .zip file and extract it.
2. Double-click the extracted WavFix application to run it. WavFix is self-contained and does not require any dependencies. Advanced users may optionally choose FFmpeg as an external conversion path (see FFmpeg); FFmpeg is free but must be installed separately.
3. If for some reason you have a problem, try right-clicking and running as administrator, or check the settings of your anti-virus software. (This should not be necessary unless you have especially restrictive anti-virus software installed).

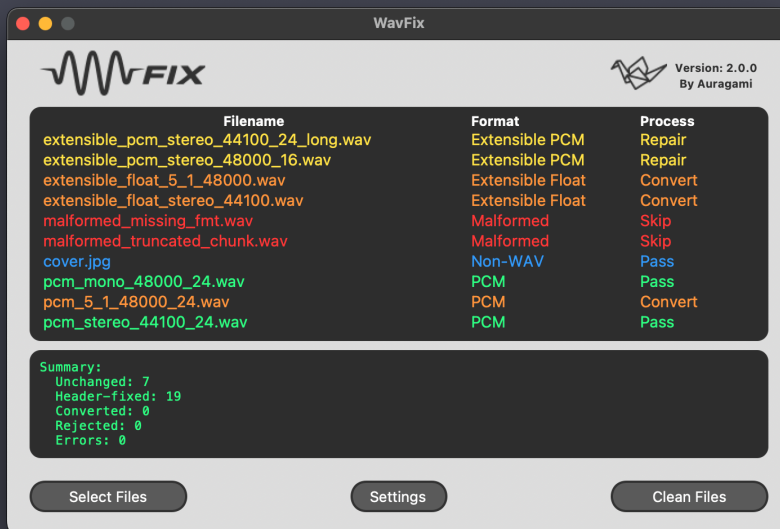
Quick Start

1. Launch WavFix.
2. Load files/folders using Select Files (or drag and drop).
3. Review the file table (Filename, Format, Process).
4. Optionally adjust settings in the Settings window.
5. Click Clean Files.
6. Choose an output folder and determine overwrite behavior if prompted.

User Interface

The WavFix user interface consists of the following components:

- A Files Window that shows the selected files including the file name, detected format, and chosen processing path.
- Select Files button to browse and select files for processing.
- Settings button to access theme, performance, and processing options.
- Clean Files button to process input and save the modified files to a selected output directory.
- An Output Window that displays file outputs.



You can drag files or folders directly into the file display to load them to WavFix.



The input and output can be cleared by double-clicking their respective windows.

Display Key

WavFix prepares audio files for Pioneer DJ hardware compatibility using a decision engine that chooses the least destructive valid path for each file.

For each file, WavFix chooses one of four processes:

- **Pass**: file is a supported non-wav file and will be copied unchanged.
- **Pass**: file is already Pioneer compatible and will be copied unchanged.
- **Repair**: extensible PCM can be normalized without changing audio data.
- **Convert**: audio data must be converted to compatible PCM.
- **Skip**: the file is unsupported, unsafe to process, or disabled by the current settings.

Further information is provided below in [Technical Details](#).

General Settings

Click the Settings button to access general application settings.

Theme

Changes preview immediately while the Settings window is open. Click Save to keep the change, or Cancel to revert to the previous theme.

Performance

Choose how aggressively WavFix uses system resources while processing.

- **Conservative**: lower system load, slower processing.
- **Balanced**: recommended default for most users.
- **Fast**: higher throughput with higher system load.

Performance changes are saved when you click Save.

Updates

Controls whether WavFix checks GitHub Releases for newer versions.

Click Check Now to manually check for a newer version. If an update is available, WavFix can open the official download page, remind you later, or skip that version.

Processing Options

Expand the Processing Options section within the Settings window to adjust how WavFix handles files.

Strictness

Controls the target compatibility profile.

- **Balanced:** preserves supported sample rates and allows compatible mono or stereo files.
- **Strict:** targets a more universal-safe Pioneer-compatible output profile.

Sample Rate

Controls what happens when a WAV file uses an unsupported sample rate.

- **Convert:** convert to the nearest supported rate.
- **Reject:** skip files with unsupported sample rates.

Supported target rates are 44.1 kHz, 48 kHz, 88.2 kHz, and 96 kHz.

Bit Depth

Controls what happens when a WAV file uses an unsupported bit depth.

- **Convert:** convert unsupported bit depths to 24-bit PCM.
- **Reject:** skip files with unsupported bit depths.

Supported PCM bit depths are 16-bit and 24-bit.

Multichannel

Controls what happens when a file has more channels than the selected profile supports.

- **Downmix:** allow WavFix to downmix multichannel audio to stereo.
- **Reject:** skip files that require multichannel conversion.

Metadata

Controls how strictly metadata is preserved during conversion.

- **Best Effort:** preserve common metadata where practical.
- **Strict Preserve:** skip conversion if metadata cannot be preserved according to policy.

Converter

Controls which engine WavFix uses when audio conversion is required.

- **Built-in:** uses WavFix's included SoundFile/libsndfile and SoXR conversion backend. This is the recommended default and does not require any additional installation.
- **FFmpeg:** uses an external FFmpeg executable for conversion. FFmpeg is free, but it must be installed separately.

When FFmpeg is selected, WavFix can either auto-detect FFmpeg from your system or use an executable selected with Choose FFmpeg.

Audio Conversion

WavFix includes a high-quality industry-standard conversion engine. It uses SoundFile/libsndfile for WAV reading and writing, SoXR for sample-rate conversion, and dithered PCM output when changing bit depth or sample format.

FFmpeg

The built-in converter is high quality and does not require FFmpeg. However, advanced users may optionally prefer to use FFmpeg as an external converter. FFmpeg is free, but it must be installed separately. If FFmpeg mode is selected within the Processing Options and WavFix cannot find FFmpeg, a warning will pop up with a download link to the official FFmpeg website.

Exporting Files

WavFix's smart export system minimizes the risk of accidentally overwriting existing files. When exporting, WavFix looks in the output destination for files that are the same type and name as whatever is at the base of its input.

The user will be asked if they wish to overwrite existing files in the chosen destination.

- If the user chooses 'Yes' - the application will overwrite any files with the same name and type in the output directory.
- If the user chooses 'No' - WavFix will create a new folder in the output directory with the same name as the input folder, appending "_clean" to the folder name.

IT IS RECOMMENDED THAT YOU DO NOT OVERWRITE YOUR ORIGINAL FILES.

Supported File Types

WavFix supports the following file types:

- Audio: .wav, .mp3, .aiff, .flac, .m4a, .ogg, .raw
- Images: .jpg, .jpeg, .png
- Documents: .txt, .pdf

Unsupported file types are ignored during selection and folder scanning. Non-WAV supported files are copied unchanged.

Technical Details

The following is a description of the program backend and the decision engine flow for users who are curious about what exactly WavFix does at a technical level.

When a WAV file is loaded, WavFix parses the RIFF/WAVE container and reads the file's chunk structure, including the fmt chunk and data chunk.

For standard PCM WAV files, WavFix leaves the file unchanged when the parsed audio format is already compatible with the selected Processing Options. If the file is PCM but has an unsupported sample rate, unsupported bit depth, or unsupported channel configuration, WavFix either converts or skips the file depending on the selected settings.

For extensible format PCM files, WavFix verifies the SubFormat GUID before making any repair. If the subtype is confirmed to be PCM and the audio data is otherwise compatible, WavFix performs a header-only repair by rewriting the WAV as canonical PCM. The audio data is copied unchanged.

For standard IEEE float WAV files and extensible float WAV files, we cannot simply relabel the file as PCM. Float audio requires true audio conversion. When conversion is approved, WavFix decodes the audio, converts it to PCM, applies sample-rate conversion only when required, and writes the result as compatible 16-bit or 24-bit PCM. Unsupported bit depths are converted to 24-bit PCM unless the Bit Depth setting is set to Reject.

When sample-rate conversion is required, WavFix converts to the nearest supported Pioneer-compatible rate: 44.1 kHz, 48 kHz, 88.2 kHz, or 96 kHz. If two supported rates are equally close, WavFix chooses the lower rate.

When multichannel audio is encountered, WavFix either skips the file or downmixes it to stereo depending on the Multichannel setting. Downmixing uses the parsed channel layout when available.

For conversion without FFmpeg, WavFix uses an in-process audio backend based on SoundFile and SoXR. Conversion is performed in streaming blocks so large files do not need to be loaded entirely into memory. WavFix uses high-quality resampling and dithered PCM quantization when reducing or changing sample format.

Metadata handling depends on the Metadata setting. In Best Effort mode, WavFix preserves common metadata where practical. In Strict Preserve mode, WavFix skips conversion when required metadata cannot be preserved according to policy.

After repair or conversion, WavFix reparses the output file to confirm that the result is a valid WAV file and that the expected action was performed safely. Header repairs are also checked to confirm that audio data was not changed.

Files that cannot be safely classified, repaired, converted, or validated are skipped.

Troubleshooting

If you encounter any issues while using WavFix, refer to the common troubleshooting steps below:

1. If you encounter an error while reading a file, ensure that the file is not corrupted or locked by another application.
2. Make sure that you have followed the installation steps correctly, especially if you are using a Mac.
3. For large directories, allow extra time. If the directory is on an external drive or thumb drive, copying it to the computer first may improve performance.
4. For VERY large directories; try inputting the subfolders from the main directory individually.
5. If WavFix reports that a file was skipped, check the Processing Options. The file may require conversion, downmixing, or metadata handling that is currently set to Reject.

Support

For additional support or to report bugs, contact me at: wavfix.dev@gmail.com

License

WavFix is licensed under the GNU General Public License 3.0 - see the license file that came with WavFix for more details.

